

Designing a Capacitive Sensor

Using Comsol Multiphysics | Project Report

The objective of this project is to design a simple capacitive sensor by using COMSOL Multiphysics software to design and test the 3D structure of the suggested capacitive sensor. The software can predict the base capacitance and hence the sensitivity of the device.

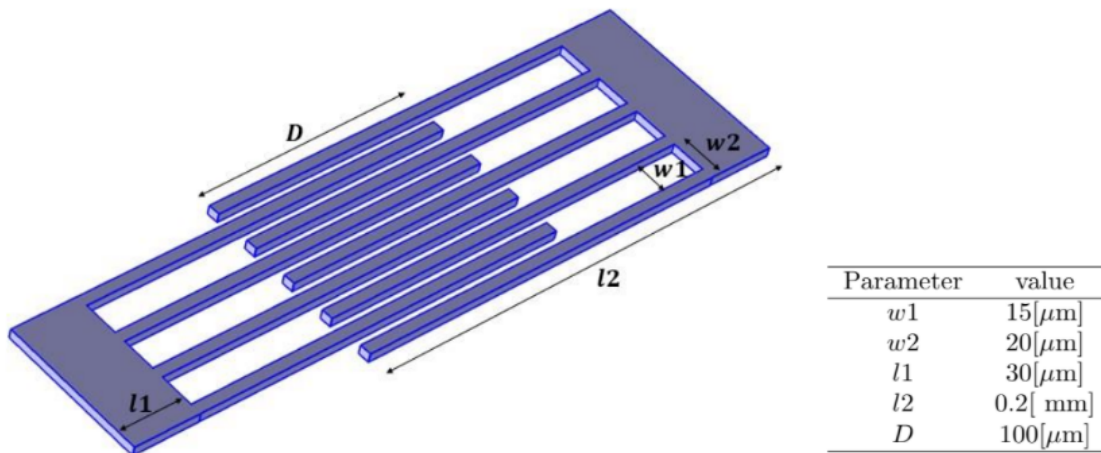


Fig.1. Comb-Drive Geometry

The applied voltage on the right arm is 4[V] and the other arm is grounded. The thickness of the comp-drive will be 4[μm] and the thickness of the substrate is taken to be 10[μm]. Through the Electrostatics module, we can solve the Poisson equation, but we need to specify the dielectric permittivities as follows:

Domain	ϵ_r
Comb-drive	11.9
Substrate	11.9
Cladding	1

Fig.2. Dielectric Permittiviteies.

Theoretical Background

Capacitors are essentially two conductors separated by a non-conductive material called a dielectric. A voltage is applied to one plate and the other plate is held at low potential or ground. This creates an electric field between the two plates. Normally, capacitors are designed to

maximize the mutual capacitance between the two plates and minimize any stray electric field lines. Compared to a standard capacitor, a capacitive sensor is designed to maximize the fringe fields between closely spaced conductors. Fringe fields loop away from the plane of the conductors as they connect one to the other. This feature allows the capacitive sensor to detect nearby objects by their effect on the electrical field created by the sensor.

Comb-drive capacitors take advantage of capacitance generated from sidewalls of electrodes. The two sets of the electrodes as the two in the Fig. 1 are placed in the same planer and parallel to the substrate. One of the two sets will be fixed and the other is let to be free to move. When talking about comb-drives, the capacitance for a single finger is just like parallel-plate capacitors,

$$C = \frac{\epsilon_0 \epsilon_r D t}{g} \quad (1)$$

In many electrostatic actuators, the gap between the electrodes is not negligible once compares to the lateral dimensions of the capacitor. Hence, fringing fields become a considerable factor and must be accounted for when modeling the electrostatic forces. The capacitance between a pair of an electrode's fingers is contributed by two things: the vertical surfaces of the finger in the overlapped region, and the fringe capacitance fields. We will be using the Palmer formula for the fringe capacitance of a rectangular parallel-plate structure,

$$C_f = \frac{\epsilon_0}{g} W L \cdot \left[1 + \frac{g}{\pi W} \left(1 + \ln \left(\frac{2\pi W}{g} \right) \right) \right] \cdot \left[1 + \frac{g}{\pi L} \left(1 + \ln \left(\frac{2\pi L}{g} \right) \right) \right] \quad (2)$$

Capacitances resulting from multiple finger pairs are connected in parallel. So, the total capacitance is a summation of the capacitance contributed by neighboring fingers,

$$C_{total} = N \cdot \frac{\epsilon_0 \epsilon_r D t}{g} + C_f \quad (3)$$

t: thickness of the comb drive, *g*: gap between the fingers of the two electrodes

W: width of the comb drive, *L*: length of the comb drive, *N*: Number of comb fingers

Software Simulation

Fig.3 shows the structure of the drive comb built in Comsol with the dectiated dimensions.

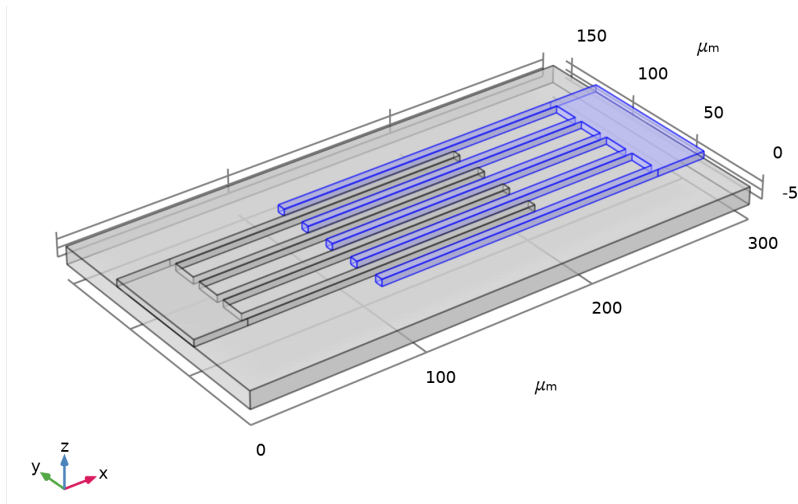


Fig.3. The geometry of MEMS comb drive.

Part 1 | Capacitance VS D

We have calculated the capacitance VS the parameter D as it ranged from $30\text{ }\mu\text{m}$ to $100\text{ }\mu\text{m}$ with a step of $2\text{ }\mu\text{m}$. A comparison between the theoretical values calculated earlier and the simulation values is made. Find the specific values in the appendix. Fig. 4 represents the change in capacitance with respect to D . *The parametric sweep was done twice: once while changing the substrate's dimensions with D , and the other while fixating the substrate's dimensions while moving the comb.*

As D gets higher, the two electrodes get closer (the gap between them is smaller). The capacitance, in this case, **increases linearly** with D which is expected from equation (3). Capacitance is **inversely proportional** to the gap between the electrodes. This is expected as the capacitances derived from multiple finger pairs are connected in parallel. The simulated results while fixating the substrate's dimensions are very close to the theoretical calculations.

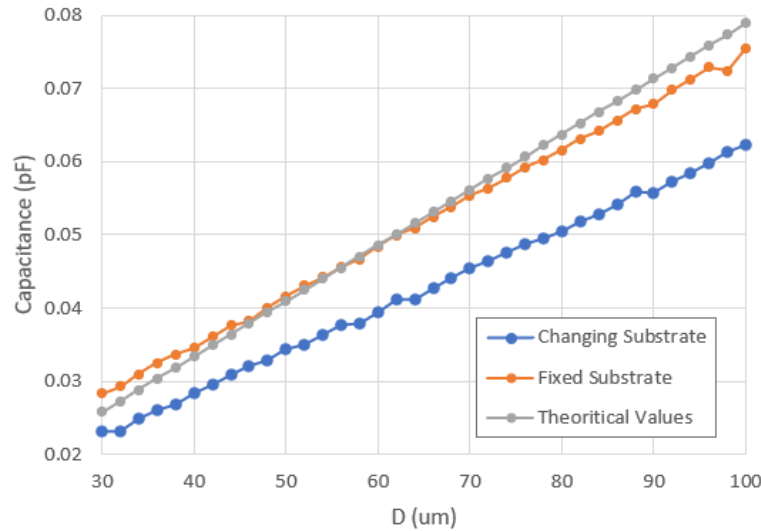


Fig.4. Capacitance (pF) VS D (μm).

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Part 2 | Electric Potential and Electric Field

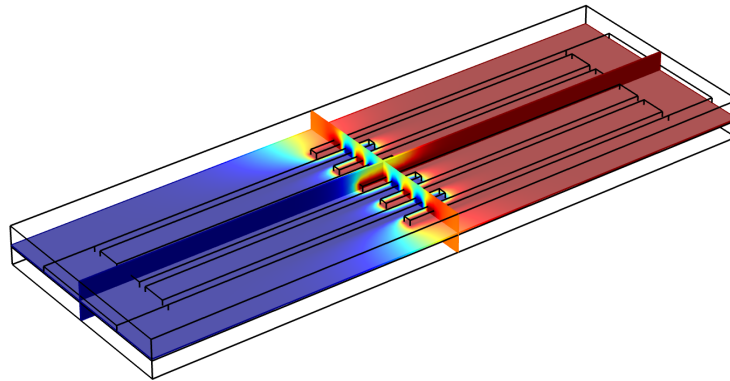


Fig.5. Electric potential distributions in the model.

The red in Fig. 5 represents high voltage [4V] and the blue shows low voltage [ground]. Then, we have Fig.6 showing the electric field (x and y components) and only the y-component of the electric field arrows (black). A qualitative depiction of the electric field lines between the fingers of a comb-drive is shown. There are different types of fields that have a distinct impact on the sensor. There are field lines that connect the vertical sides of the fingers and those that connect the top sides of the fingers. Those larger arrows (strong E) are associated with a component of the force that tends to align the fingers. Then, There are other arrows connecting the distant two top sides of the electrodes (weak E). Those tend to disengage the fingers. Lastly,

there are field lines connecting the tips of the finger with the sides of its neighbouring finger (small arrows going to the side). Those do not contribute to the force.

Fig. 7 also shows the electric field's streamlines and their direction in comparison with the potential changes across the combs.

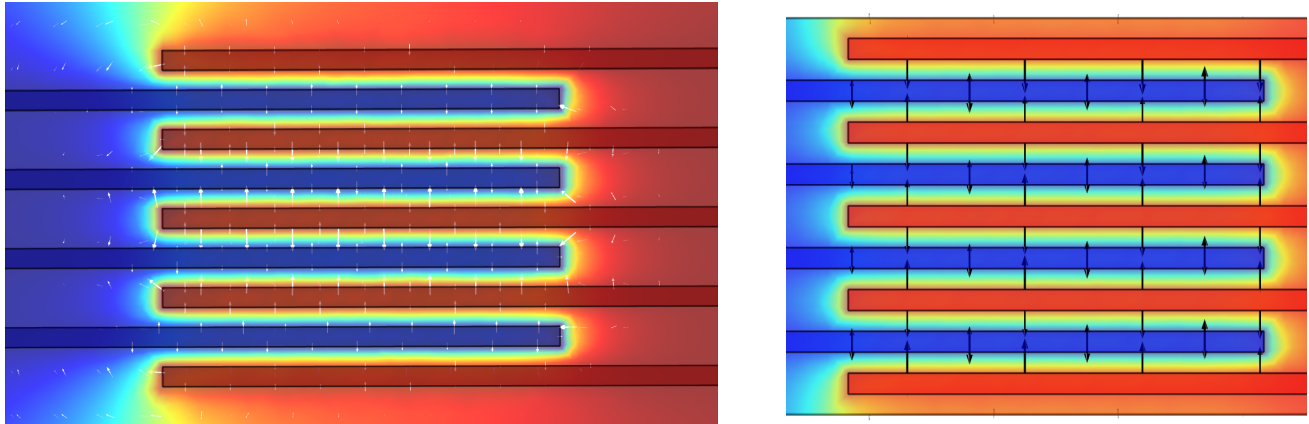


Fig.6. Electric Field: all components (Left) and Only y-component (Right).

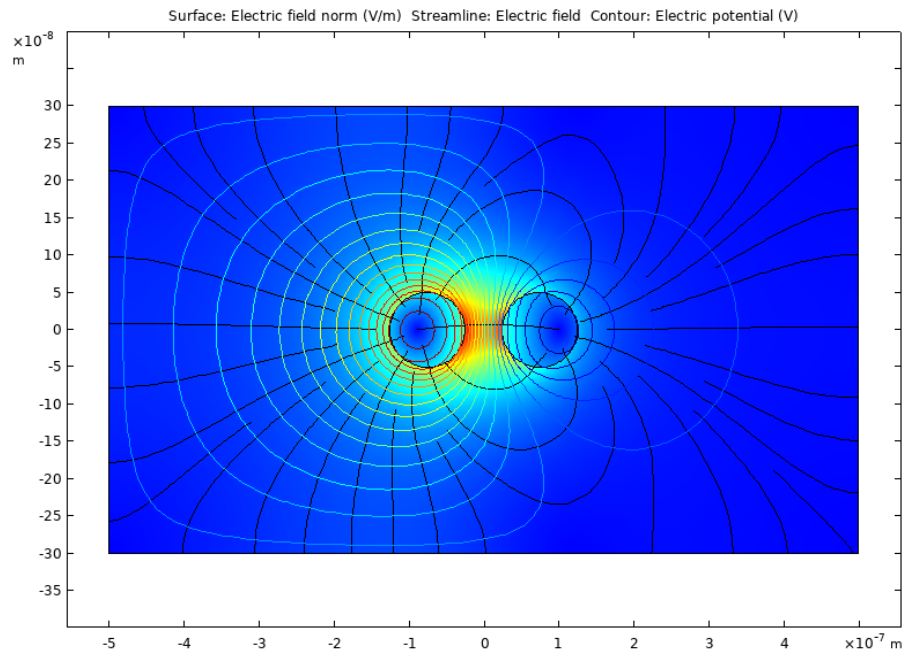


Fig.7. Normal Electric Field and the Electric Field's Streamlines.

Lastly, we plot the relationship between the electric potential and the y-component of the electric field at $D = 100 \mu\text{m}$. As expected, it shows that the relation between them is quadratic.

$$E = \text{constant} * V^2.$$

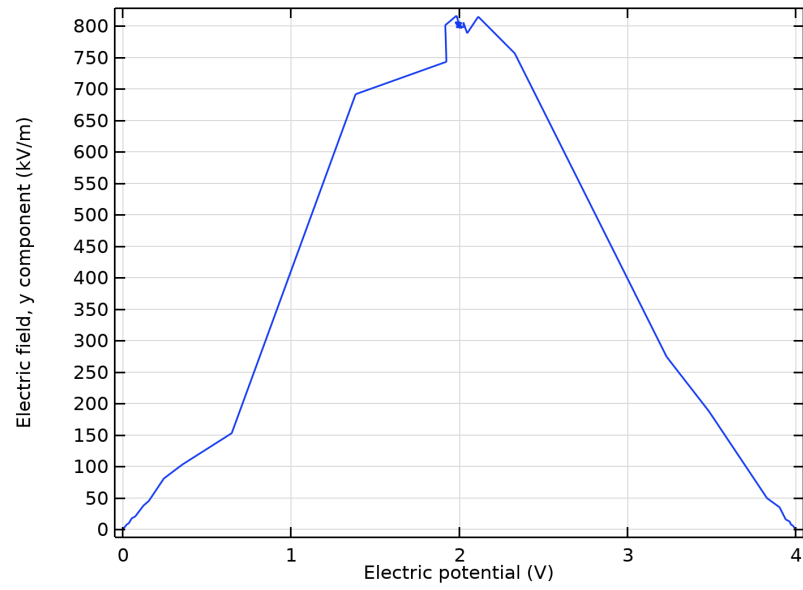


Fig.7. Applied Electric Potential Vs E_y

Appendix

D (um)	Theoretical Capacitance (pF)	Simulated Capacitance (pF) – Moving Substrate	Simulated Capacitance (pF) – Fixed Substrate
30	0.025794	0.02313	0.02829
32	0.027312	0.023074	0.029285
34	0.028829	0.024984	0.030969
36	0.030346	0.026105	0.032457
38	0.031863	0.026921	0.033696
40	0.033381	0.028289	0.034566
42	0.034898	0.029534	0.036113
44	0.036415	0.030963	0.03763
46	0.037932	0.031984	0.038194
48	0.03945	0.032825	0.040071
50	0.040967	0.034418	0.041605
52	0.042484	0.035032	0.043056
54	0.044001	0.036259	0.044246
56	0.045519	0.037608	0.045584
58	0.047036	0.03792	0.046681
60	0.048553	0.039395	0.048342
62	0.05007	0.041105	0.04993
64	0.051588	0.041164	0.050886
66	0.053105	0.042715	0.052413
68	0.054622	0.044112	0.053847
70	0.056139	0.045432	0.055406
72	0.057657	0.046393	0.056311
74	0.059174	0.04763	0.057834
76	0.060691	0.048766	0.059249
78	0.062208	0.049446	0.060247

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80	0.063726	0.050518	0.061548
82	0.065243	0.051918	0.063101
84	0.06676	0.05275	0.064209
86	0.068278	0.05417	0.065666
88	0.069795	0.055979	0.067219
90	0.071312	0.055745	0.067886
92	0.072829	0.057343	0.069802
94	0.074347	0.058472	0.071244
96	0.075864	0.059804	0.072938
98	0.077381	0.061299	0.07242
100	0.078898	0.062303	0.07545